

VALID: A 12-Item Validation Checklist for Financial Machine Learning

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SSRN 6508779



GitHub repo

Statistically perfect. Economically worthless.

340 strategy variants · 80 papers audited · one 12-item gate that tells them apart in ~5 minutes

The Field Today

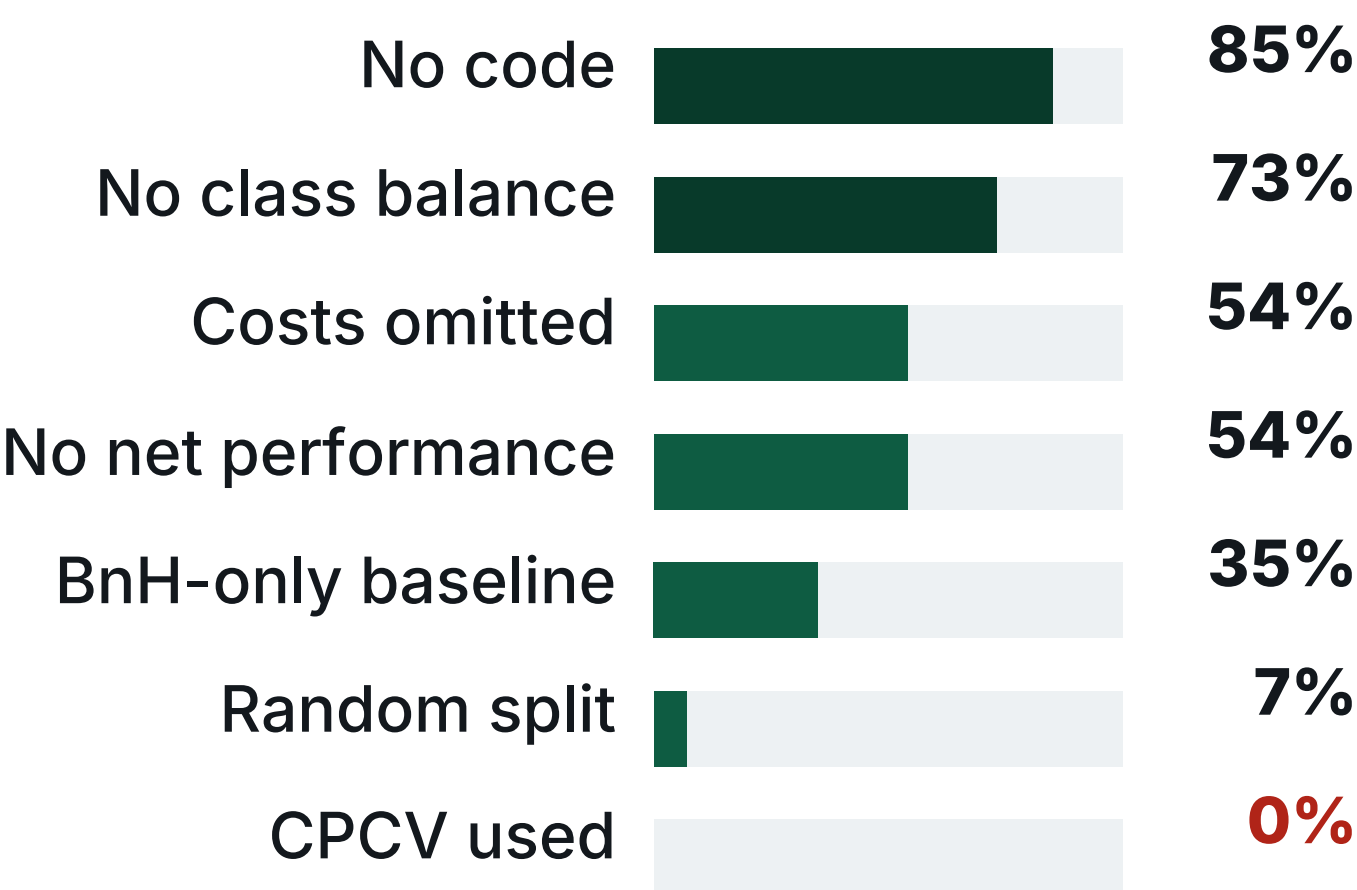
2.5

of 12

items met

median audited paper (n = 74 empirical, 2018–2026)

Share of papers failing each dimension



This is the reporting norm, not a few bad papers.

The Strategies That Passed Everything

PASSED

9 of 340 variants

Bonferroni: 9

Holm 9 · BH-FDR 10

t up to 21.7

Harvey t > 3.0: 10 survive

SR up to 1.98

vs benchmark 0.917

Every classical correction: PASS

Multiple-testing control is necessary — and demonstrably not sufficient.

Across all 340 variants: 52% lose money after costs · only 4.4% beat the benchmark.

All nine trade daily. CPCV rates every one fully overfit; the DSR rejects all 340.

REALITY

the same nine

PBO = 1.0

all nine — complete overfitting

AUC 0.47–0.64

near random

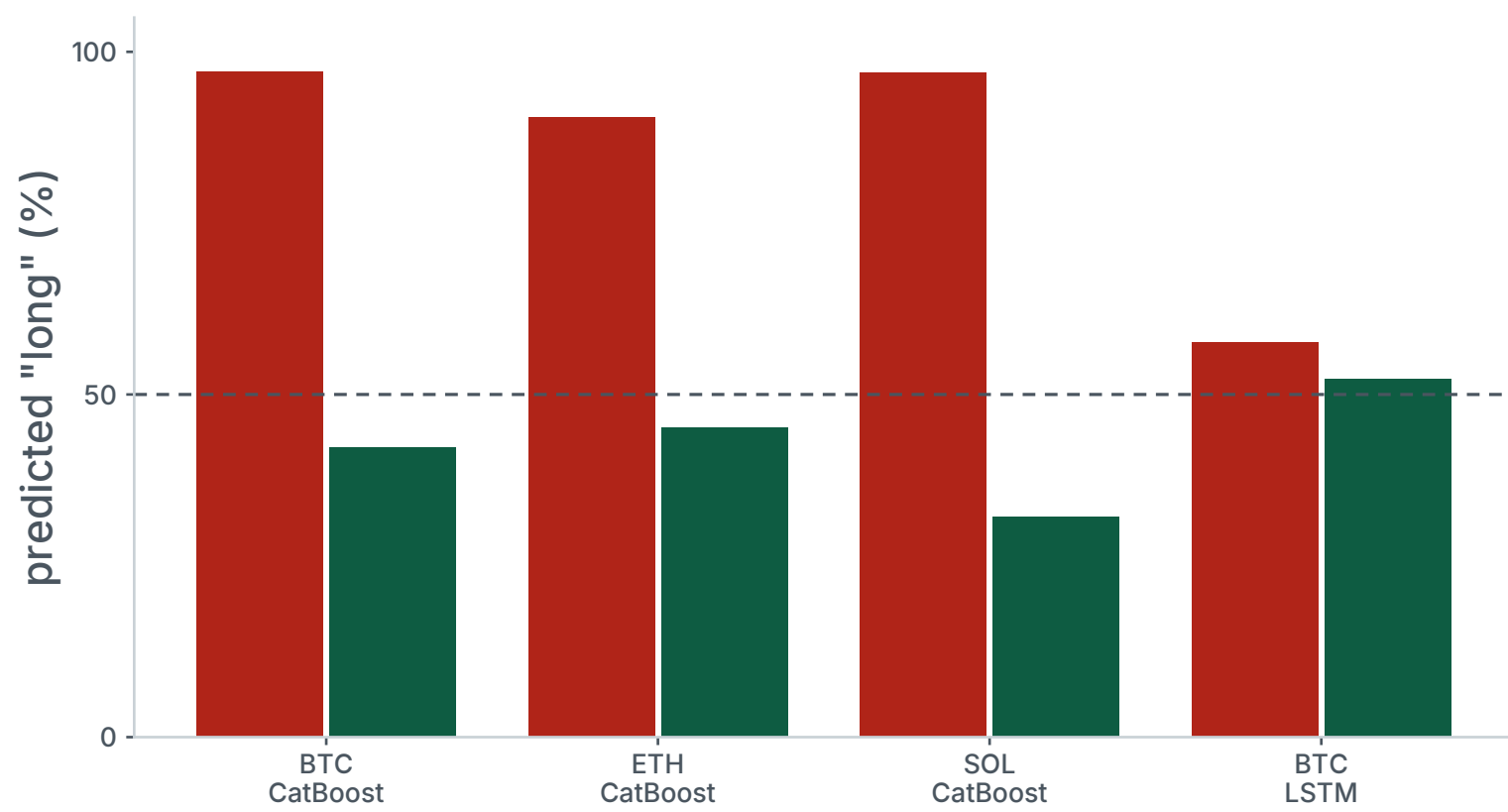
DSR: 0 of 340

E[max SR] under null 2.93

Corrections cannot see overfitting

Bull Bias

structural class imbalance (V1–V2)



Tree models predict “long” 90–97% of the time. After balancing, AUC converges to ≈ 0.50 — a coin flip.

Balanced AUC (BTC / ETH / SOL) 0.50 · 0.49 · 0.51

Net SR, 1h balanced -0.88 · -1.28 · -0.21

All 1h models: PBO = 1.000.

The Gate

Stage 1 — Statistical

V1–V6 · balance, temporal split, CPCV+PBO, variance, permutation



Stage 2 — Economic

V7–V12 · net costs, cost sweep, baselines, regimes, turnover, code

Binary items · fixed order · pass/fail

False positives on signal-free data (n = 200)

AUC alone 27% [21, 34]

+ CPCV & PBO 0% [0, 1.9]

One item (V4) does it — used by 0 of 74.

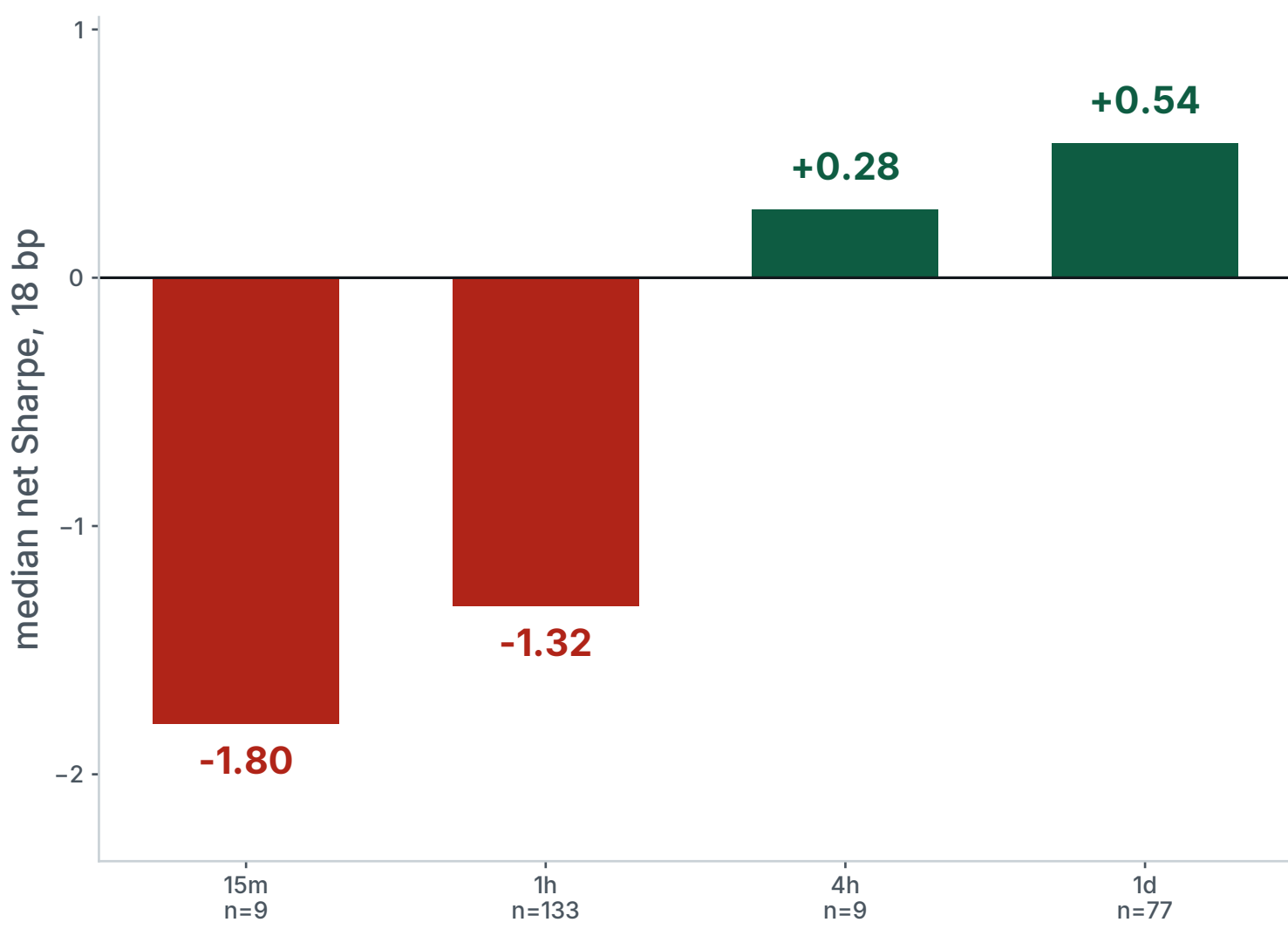


Run the checklist on your own strategy ~5 minutes

orcajae.github.io/valid-framework

Cost Illusion

net-of-cost performance by frequency (V7–V8, V11)



	median	best	< 0
15m (n = 9)	-1.80	-1.33	100%
1h (n = 133)	-1.32	+0.32	96%
4h (n = 9)	+0.28	+0.57	44%
1d (n = 77)	+0.54	+1.98	22%

One 1h CatBoost run: gross SR 0.750 → net 0.135 at 18 bp

— 82% of gross Sharpe consumed at 101 trades / yr.

340-variant corpus, 18 bp round-trip, model variants only.

Score the last paper you reviewed

□□□□□□□□□□□□

median in our audit: 2.5 / 12

☐ V1 Report prediction class distribution

☐ V2 Test with / without class balancing

☐ V3 Use temporal splitting only

☐ V4 Apply CPCV with PBO

☐ V5 Report Var(SR_IS)

☐ V6 Include permutation tests (≥ 100)

☐ V7 Report net performance with costs

☐ V8 Cost sensitivity analysis

☐ V9 Compare against simple baselines

☐ V10 Evaluate in bear markets and regime transitions

☐ V11 Report trade frequency

☐ V12 Provide code for reproducibility

V1–V6 Stage 1 (statistical)

V7–V12 Stage 2 (economic)

Self-audit of this paper

9 / 12

fully satisfied

Three partials disclosed: V5 (flatness criteria disagree), V8 (single 18 bp cost level), V10 (3 crypto assets only).

v1.0 — not a certification. The checklist has had no formal Delphi review; community revision is invited.

Who Is This For?

adoption paths

Reviewers & PCs

A minimum reporting standard for financial ML submissions.

Validation & model-risk teams

A two-stage gate for incoming strategies, both binary.

Allocators & operational due diligence

Twelve questions before trusting a pitched Sharpe ratio.

Open source · MIT license

Coding sheet for all 80 papers and per-paper cells are public.

Selected references

Harvey, Liu & Zhu (2016). ...and the Cross-Section of Expected Returns. RFS 29(1).

McLean & Pontiff (2016). Does Academic Research Destroy Return Predictability? JF 71(1).

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